

Mechanical Gears

Student Edition — VEX EXP · C++ in VS Code

Name: _____ Date: _____ Period: _____

Introduction

Gears trade speed for torque — and this chapter pushes that idea three steps further. An **idler gear** placed between the drive and driven gears changes the direction but *not* the ratio — only the first and last gears count. To reach a *big* ratio, you **compound**: two gears share a shaft so the stages multiply (two 2.5:1 stages give 6.25:1). And not every gear is a flat spur gear — your kit also has **bevel** gears (turn a 90° corner), a **worm & wheel** (huge one-step reduction that won't back-drive), **rack & pinion** (rotary becomes straight-line motion), and **sprockets & chain** (send power across a distance). In this activity you'll build gear trains, verify ratios by counting turns, and chart what each mechanism can do. Read Chapter 4 in the textbook for the full explanation.

Learning Objectives

- Calculate a gear ratio from tooth counts and predict whether a train gears up or gears down.
- Explain why an idler gear changes direction but not the ratio.
- Build a compound gear train and find its overall ratio by multiplying the stages.
- Identify the five gear/sprocket types in the EXP kit and state what each is for.
- Verify a predicted ratio by counting turns of the drive and driven shafts.

Key Ideas (quick reference)

- **Idlers don't change the ratio.** Only the first (drive) and last (driven) gears set the ratio; any gears between just pass motion along and cancel out. Use an idler to bridge a gap or flip direction. Odd # of gears in a line → same direction; even → opposite.
- **Compound = multiply the stages.** When two gears share a locked shaft, the middle counts DON'T cancel. Total ratio = stage 1 × stage 2. Two 2.5:1 stages → 6.25:1. This is how you beat the kit's 2.5:1 single-stage limit.
- **Five types, five jobs.** Spur (everyday, one plane) · Sprocket & chain (power across a distance, same direction) · Bevel (90° corner) · Worm & wheel (huge reduction, self-locks, won't back-drive) · Rack & pinion (rotary → linear).
- **The worm's superpower.** A single-start worm turns a 24-tooth wheel one tooth per turn = 24:1 from one pair, and the wheel can't drive the worm back — so a worm-driven lift holds its height with the motor off.

Build Constraints

Build constraints (all year): (1) Build from your classroom Gateway kit. It's built around EXP but also bundles some V5 mechanism parts (the Smart Motor, sensors, the Winch & Pulley Kit) — those are yours to use. What's off-limits is the separate competition team's V5 kit: don't pull from it for class builds, so it stays complete for the team. (2) Your whole build must fit one storage cubby — an IKEA Kallax shelf compartment: a 33 × 33 cm (13 × 13 in) opening, about 38 cm (15 in) deep. The 33 × 33 cm face is the binding limit (the build must pass through it). Design compact and break builds down to store them.

Equipment & Materials

- VEX EXP gears (24T, 36T, 48T, 60T), at least one worm + worm wheel, bevel gears, a rack + pinion, sprockets + chain
- Shafts, bearing flats, shaft collars, spacers, and a C-channel or flat-bar baseplate
- A hand crank (or a free shaft to turn by hand); tape to mark a tooth
- Engineering notebook; optional: one EXP Smart Motor to drive a train

Procedure

1. **Inventory your gears.** Lay out the 24/36/48/60T spur gears and count teeth to confirm the numbers.
2. **Build a simple pair and verify it.** Mesh a small drive gear with a larger driven gear. Tape-mark a tooth on each, turn the drive exactly 5 full turns, and count the driven turns. Does turns-in $\div$ turns-out match the ratio from tooth counts?
3. **Add an idler — watch it not matter.** Insert a third gear between them. Predict the new ratio (trick question), then re-run the 5-turn test. Note the new output direction.
4. **Compound it.** Build a two-stage train with a shared middle shaft. Predict the overall ratio by multiplying the stages, then measure by counting turns. Record predicted vs. measured.
5. **Meet the special gears.** Build or examine a worm & wheel (try to back-drive it — you can't), a rack & pinion (rotary $\rightarrow$ linear), and a bevel pair (power turns a 90° corner).
6. **Fill in the mechanism chart** below for each mechanism you built.

Compound Gear-Train Practice

Each stage = **driven : drive**. Overall ratio = stage 1 $\times$ stage 2. First row is worked for you.

Stage 1 (drive $\rightarrow$ driven)	Stage 2 (drive $\rightarrow$ driven)	Overall ratio	Speed & torque effect
24T $\rightarrow$ 48T (2:1)	24T $\rightarrow$ 60T (2.5:1)	$2 \times 2.5 = 5:1$	Output 5 $\times$ slower, ~5 $\times$ torque. Stages needn't match — multiply them.
24T $\rightarrow$ 60T	24T $\rightarrow$ 60T		
36T $\rightarrow$ 60T	24T $\rightarrow$ 48T		
single 24T $\rightarrow$ 60T	— (none)		

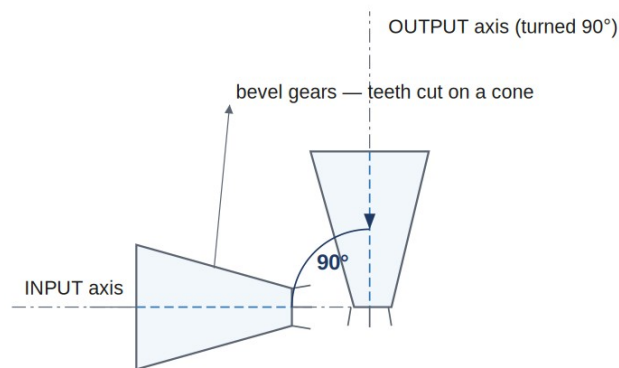
Mechanism Chart

For each mechanism, mark Y or N. The first row is filled in as a model.

Mechanism	↑ Speed?	↑ Torque?	Reversible?	Changes motion type?	Real-life example
Simple gear train	Y	Y	Y	N	Drivetrains, clocks
Simple gear train with idler					
Bevel gear					

Mechanism	↑ Speed?	↑ Torque?	Reversible?	Changes motion type?	Real-life example
Worm & wheel					
Rack & pinion					
Sprocket & chain					
Belt & pulley					

Engineering Drawing



BEVEL GEAR PAIR — 90° DRIVE	
FUNCTION: turn the axis of motion 90°	
UNITS: inches	DRAWN BY:
SCALE: not to scale	DATE:

A bevel gear pair turns the axis of motion 90° — the windmill's key mechanism in the next chapter.

Code It — One Line, Timed

In Chapter 3 you started and stopped a motor by hand. **spinFor** does a timed run in one line — the motor stops on its own. In the starter's marked region, type:

Type this in the marked region

```
leftMotor.setVelocity(50, percent);
leftMotor.spinFor(forward, 3, seconds);
```

Now try: build a compound train (the multiply rule) and run it for 3 seconds. A big reduction barely creeps — but with far more force; geared up it whirls. Change 3 to 6 to go twice as far. The compounded ratio, not the code, sets how far the output turns.

Conclusion Questions

Answer in complete sentences.

1. A 12-tooth driver turns a 36-tooth driven gear. What is the gear ratio, and does this gear up or down?
2. You need a robot arm to lift a heavy load slowly and steadily. Gear up or gear down?

3. A 24T gear drives a 48T idler, which drives a 60T gear. What is the overall ratio — and what did the idler change?
4. Build a compound train from your kit's gears that reaches a reduction of at least 6:1. List the gears in each stage and show the multiplication.
5. Your robot's arm needs to hold its position when the motor is off, or it sags. Which gear type solves this, and why?
6. You want a claw that opens and closes in a straight line. Which mechanism converts the motor's rotary motion into that linear motion?